

Business

Which aspects of your academic background do you believe best demonstrate your preparation for this research area? Examples may include, but are not limited to: courses completed, independent study, test scores, etc. (100 words)

Can you share your final grades for below? Or mid-term results if the below is work in progress?

The below best demonstrates my preparation for Business:

1. Completed Wharton Global Youth program Essentials of Finance
2. Accepted in 123 Entrepreneurship and Social Innovation Program (HESIP)
3. AMC 12 B - 97.5 score
4. National Economics Challenge - National Finals 2025
5. Wharton Investment Competition - First Placement

Do you have any other experience or qualifications in this field? Tell us anything about your background that may be helpful in demonstrating your interest in this specific subject area. (100 words)

I worked as an intern and volunteer at XYZ, Shanghai for 3 summers teaching underprivileged Chinese kids in personal finance. As an intern, I helped the marketing team re-brand XYZ's image and fundraised for education accessibility. I am also the head of NW, a NH chapter that teaches math and tennis to elementary students. From these leadership roles, I gained pivotal experience in running a non-profit enterprise. Similar to profit oriented organizations, communication and alignment with partners is critical. Currently, I am also participating in the Diamond Business Challenge and a member of 123 FBLA.

Pioneer considers both your specific interests and broad intellectual curiosity within each research area. Please tell us why you want to engage in this research area, clarifying your openness to exploring beyond the interests you describe (100 words minimum, 150 words maximum).

Businesses affect everything from global markets to problem-solving innovations. They shape the flow of resources, influence consumers, and drive competition that generates new ideas. Businesses are also the driving force of societal impact. In the summer of my freshman year, I founded the FinTech Initiative, a non-profit program where high school students volunteer to teach financial and digital literacy at elementary schools around the NH area. Running this program proved similar to the management of a business, which I experienced in my summers at XYZ. I was able to see firsthand the impact of my organization in my own community and when imagining it on a larger scale, I can envision the change businesses can cause in the world. To create impact at this scale, I aim to study how ventures grow into major companies by understanding every aspect of business, from financial decisions to operational strategy.

Engineering

Which aspects of your academic background do you believe best demonstrate your preparation for this research area? Examples may include, but are not limited to: courses completed, independent study, test scores, etc.

The below best demonstrates my preparation for Engineering:

1. Completed Wharton Global Youth program Essentials of Finance
2. Precalculus Accelerated - A
3. AP Calculus BC - A grade and 5 AP score
4. AP Physics 1 - A grade and 5 AP score
5. AMC 12 B - 97.5 score
6. Linear Algebra - A+ midterm
7. AP Physics C - A midterm
8. Independent Study: Python, Java, C++, and code algorithms
9. National Economic Challenge - National Finals 2025
10. Independent Study: Micro and macro economics

Do you have any other experience or qualifications in this field? Tell us anything about your background that may be helpful in demonstrating your interest in this specific subject area.

I am interested in financial engineering. I am on the Wharton Investment Competition team where we have created a strategy for long-term investing and methods to navigate the stock market. On the team, I handle most of the research of specific stocks we are considering, including both qualitative and quantitative analysis. My team for the National Economics Challenge in 2025 reached the National Finals in Atlanta and we learned to develop solutions to real-world economic issues. I am also the CFO for my school's Diamond Challenge team, developing strategic financial methods to support our business idea.

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I have been an avid investor in the stock market, in simulators and competitions. The market is an important factor in global and national economies, influencing every industry. Everyday, there are numerous factors that influence movement in the market, creating volatility that is almost impossible to predict. Yet, corporations and funds have created algorithms and systems that hedge risk and optimize portfolios. I find it fascinating that quantitative tools and models can turn the irregularity of financial markets into structured strategies. I want to understand not only how these mechanisms work, but why they work. In my research, I want to examine these designs more deeply and eventually create my own that will help others learn how to navigate the volatility of the stock market.

Application Essay

"Often the right path is the one that may be hardest for you to follow. But the hard path is also the one that will make you grow as a human being." – Karen Mueller Coombs

The Pioneer Research Institute is widely recognized as one of the most rigorous and challenging academic opportunities available to high school students. Reflecting on the quote above, discuss **how Pioneer's high academic standards will help you reach your full potential and achieve your goals.**

In your response, please directly reference and engage with the quote above. Clearly explain **why you have chosen your academic goals** and **how specific features of the Pioneer Research Institute will support your growth.**

We encourage you to **incorporate ideas from Pioneer's website** or share insights you've gained from others familiar with the Institute. Your response should be between 300 and 400 words (300 words minimum, 400 words maximum). Please note that using AI is not allowed; the prompt and all essays will be checked for AI generation.

Karen Mueller Coombs quote rivets me as I am a proponent of the growth mindset which is transformative. As such, I am aligned with Pioneer's rigor as it actualizes this mindset – through hard work and perseverance, in-depth research and focus; there is human transformation.

My academic goals are driven by curiosity and the desire to take knowledge and transform it into societal impact. One of my pledges is to ensure education for all, especially in the context of financial literacy as this is the foundation of family stability. To reach this goal, I must pursue academic scenarios that stretch me beyond what I know and this is what I believe Pioneer can do for me given its professor mentorship model. Although my school teachers are solid mentors, they don't have experiences or expertise that distinguished scholars have. In the Pioneer Spotlight, scholar Jada '22, spoke about the importance of her one-on-one class with her professor who was always there to pick her up and support her throughout her entire journey. I look forward to such academic mentorship.

I have already seen the benefit of pushing through difficulty which leads to transformative growth. Alongside taking some of the most rigorous courses at my school, I searched for opportunities outside of class to extend myself in the areas that interest me and joined my school's first Wharton Investment Competition team. Since this was our first time, we struggled with tasks and rules. As the one with the most experience in the stock market, I took the initiative to delve deeper into the competition including the case studies, regulations, and deliverables. By sharing my knowledge and organizing meetings, I was able to guide our team in building a strong portfolio and submit a solid midterm report. The experience was not easy yet I appreciated and valued not only the new finance lessons I learned but also the heartier resilience and the honing of my leadership skills.

Coombs shares that the right path is the one that may be the hardest for me to follow. Yet I am up for the challenge and look forward to Pioneer's rigorous challenges as I know that once I overcome, the other side is transformative growth which I look forward to sharing with the rest of the world in the context of global financial stability.